Phuong-Linh Huynh-Ha

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Education

- **University of Science (HCMUS), Ho Chi Minh City, Vietnam** Sep. 2022 – Present
  Advanced Program in Computer Science; GPA: 3.9/4.0
  - Granted **Merit-Based Scholarship (Top 4 GPA)** in Semester 2, 2024-2025.
  - Being one of **five recipients** of **Career Readiness Scholarship**, awarded by **Gameloft** in partnership with **HCMUS** for demonstrating strong professional potential and academic merit in 2025.
  - Achieved **Top 1.9% score on the VNU-HCMC Assessment Test** (HSA)
  - Relevant coursework: Artificial Intelligence, Computer Vision, Natural Language Processing, Information Retrieval, Computer Systems, Statistics.
- **Annual Summer School on Mathematical Aspects of Data Science** Jun. 2026 – Jul. 2026
  Institute for Mathematical Sciences, National University of Singapore
  - Selected for a competitive intensive program in mathematical data science, targeted primarily at PhD students.
  - Coursework including Random Matrix Theory, Sequential Decision-Making, Robust Statistics,...

Selected Publications

- Thanh-Khoi Nguyen, Hoang-Phuc Nguyen, **Phuong-Linh Huynh-Ha**, Minh-Triet Tran. **A Top-Down Framework for Metric-Scale Athlete Localization from Single Broadcast Frames**. *Accepted for ICMV 2026.*

  Co-authored a monocular athlete localization framework introducing **Boundary-Aware Adaptive Tiling** and a specialized RTMPose-based keypoint estimator for world-coordinate localization, achieving **97.44 LocSim** and **0.9128 mAP** on the SoccerNet SynLoc benchmark.
- Hoang-Phuc Nguyen, **Phuong-Linh Huynh-Ha**, Minh-Triet Tran. **Generalizability Evaluation and Anchor-Guided Approach for Category-Agnostic Pose Estimation**. *SOICT 2025, Springer CCIS.*

  Co-authored and introduced a **training-free anchor-guided geometric mapping method** based on Delaunay triangulation, improving fine-grained keypoint accuracy and mitigating overfitting in category-agnostic pose estimation.
- Hoang-Phuc Nguyen, **Phuong-Linh Huynh-Ha**, Hai-Dang Nguyen, Minh-Triet Tran. **Endo-TBR: Temporal Basis Reallocation for Efficient Dynamic Gaussian Splatting in Surgical Scenes**. *Under Review for ICONIP 2026.*

  Co-authored a **temporal basis reallocation strategy** for dynamic Gaussian splatting that reallocates low-contribution radial basis functions to better capture localized tissue motion, improving reconstruction fidelity while maintaining rendering efficiency.

Research Experience

- **KIT Computer Vision Lab – Kyoto Institute of Technology (KIT)** Jan. 2026 – Mar. 2026
  Research Intern, supervised by Prof. Nobuhara Shohei
  - Developed a comparative evaluation framework for **optical flow** and **2D correspondence estimation** methods in **near-rotational** and **low-parallax environments**.
  - Analyzed motion ambiguity in **feature-poor scenes** and investigated the use of correspondence cues to improve SLAM-based pose estimation.
- **Software Engineering Laboratory – SELab HCMUS** Aug. 2025 – Present
  Thesis student, supervised by Assoc. Prof Minh-Triet Tran on Computer Vision
  - Conducting research in computer vision, focusing on **category-agnostic pose estimation** and **dynamic surgical scene reconstruction**.
  - Developing EndoFGS, **a frequency-domain Gaussian splatting framework** for surgical scene reconstruction in feature-poor endoscopic environments.
  - Participated in the **CVPR 2026 Spiideo SoccerNet SynLoc Challenge**, ranked top 1; developed methods to detect and localize athletes in single-frame world coordinates utilizing synthetic data.

Achievements and Awards

- **Jensen Huang Scholarship (NVIDIA & Vietnam National University system)** Feb. 2026
  - Selected as one of **50 students** for a $1M AI talent initiative based on academic merit and research potential.
  - Participating in intensive AI training and gaining access to NVIDIA GPU clusters.
- **Artificial Intelligence Olympiad for University Students (3-person team)** Nov. 2025 – Dec. 2025
  Organized by Vietnam Association for Information Processing (VAIP)
  - Awarded **Third Prize** at the **National Finals** (Dec. 2025).
  - Awarded **First Prize**, **Champion** across **all Regional Preliminary Rounds** (Nov. 2025).
- **Odon Vallet Scholarship** Aug. 2022
  - Awarded prestigious **Odon Vallet Fellowship** for academic excellence, issued by Rencontres du Vietnam.
- **Vietnam National Olympiad in English** Apr. 2022, Feb. 2021
  - **Bronze Medal** in 2022 and **Honorable Mention** in 2021, awarded by Vietnam Ministry of Education and Training.
  - **First Place** in the Provincial Olympiad in English of Quang Ngai throughout 2021 - 2022.

Projects

- **EndoFGS: Surgical Scene Reconstruction & 3D Gaussian Splatting** Ongoing Research
  - Building a surgical scene reconstruction pipeline using 3D Gaussian Splatting, evaluated on the EndoNeRF and StereoMIS datasets.
  - Improving rendering quality (PSNR/SSIM) on textureless tissues by applying frequency-domain filters, including Wavelet, Fourier, and high-boost, to enhance image features.
- **Robust 3D Reconstruction via 2D Correspondence Integration** Ongoing Research
  - Extracting and comparing optical flow from MegaSaM and CoTracker3 to resolve motion ambiguity in low-parallax and near-rotational videos.
  - Using heatmap analysis to visualize flow differences, creating a reliable method to correct SLAM tracking errors in feature-poor environments.

Extracurricular Activities

- **Youth Union Branch Commendation**
  University of Science, VNU-HCM
  Awarded a **Certificate of Merit** in 2023 and the **Outstanding Youth** title in 2026 for contributions to student activities.
- **Inspire to Aspire Seminar Series** 2023
  Organizer – Ho Chi Minh City, Vietnam
  Co-organized a career seminar series connecting students with industry professionals.

Skills & Interests

Technical Skills:	Scientific Research, Problem Solving, Machine Learning, Computer Vision
Technical Languages:	C/C++, Python, Kotlin, SQL, LaTeX, Markdown, CSS, HTML
Frameworks/Libraries:	PyTorch, NumPy, Pandas, R, Jetpack Compose, SFML
Tools:	Google Colab, Docker, Git/GitHub, Android SDK, 3D Slicer
Languages:	English (IELTS 8.0), Vietnamese (native), Japanese (beginner)
Interests:	Cycling, Swimming, Singing, Guitar (learning), Poker, Billiard